

Dragonwing QCS6490 - NPU Benchmark Report

Hexagon V68 via QNN HTP backend - generated 2026-05-14 06:22 UTC

Test platform

SoC	Qualcomm Dragonwing QCS6490 (Thundercomm Rubik Pi 3)
NPU	Hexagon V68, accessed via QNN HTP backend on /dev/fastrpc-cdsp
CPU baseline	Kryo 670 (4x Cortex-A78 + 4x A55), 4 threads via XNNPACK
Runtime	TFLite (ai-edge-litert 2.1.4) + libQnnTFLiteDelegate.so
Host OS	Yocto / Qualcomm Linux Reference Distro 2.0, kernel 7.0.0
Quantization	w8a8 (compiled via Qualcomm AI Hub, target Dragonwing RB3 Gen 2)
Method	Random input tensors; warmup then steady-state timing

Results

Model	Input	CPU latency	CPU fps	NPU latency	NPU fps	Speedup
PoseNet-MobileNet (w8a8)	513x257x3	17.40 ms	57.5	4.04 ms	247.4	4.30x
YOLOv8-Det (w8a8)	640x640x3	31.86 ms	31.4	6.02 ms	166.0	5.29x
SSD-MobileNetV2 (fp32->q)	300x300x3	23.23 ms	43.0	5.59 ms	178.8	4.15x

Headline

The Hexagon V68 delivers a 4-5x latency improvement over a 4-thread Cortex-A78 CPU baseline. YOLOv8 object detection runs at 166 fps at 640x640 - comfortably real-time with headroom for multi-stream or higher-resolution input.

Observations

- * Thermal: SoC temperatures rose from 42-44 C to 45-46 C across ~3 minutes of continuous NPU work. No throttling observed.
- * SSD-MobileNetV2 was distributed as fp32 but HTP reported "SocModel doesn't support FP16" and fell back to an int8 path on-device. To obtain a clean fp16/w8a16 baseline, recompile via AI Hub with --precision w8a16.
- * DDR bandwidth (reported by the HTP runtime): YOLOv8 read ~4.8 MB and SSD-MobileNetV2 read ~8.4 MB per inference - useful for memory-bound analysis.
- * Per-inference CPU timing was measured under XNNPACK with num_threads=4. Higher thread counts (up to 8 with the A55 cluster) yield marginal gains on these workloads.
- * All NPU runs used 300-500 timed iterations after 20-iter warmup. Variance was <2% across repeated runs.

Reproducing

Benchmark script: /tmp/npu-bench/bench2.py on the Rubik Pi 3 host. Run inside the rubikpi3-cam-test:npu image with /dev/fastrpc-cdsp, /opt/qnn-libs and /usr/lib/dsp bind-mounted, ADSP_LIBRARY_PATH=/usr/lib/dsp, LD_LIBRARY_PATH=/opt/qnn:/usr/lib.

